MA2101- Linear Algebra via Matrices

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Lecture - 5&6

Trace

Definition

Let $A = [a_{ij}] \in \mathbb{F}^{n\times n}$ be a square matrix. Then the sum

$$tr(A) := a_{11} + a_{22} + \cdots + a_{nn}$$

is called the trace of A.

- $tr(I_n) = n$ and $tr(0) = 0$.
- $tr(\alpha A) = \alpha tr(A)$ for all $\alpha \in \mathbb{F}$.
- $tr(A^t) = tr(A)$ and $tr(A^*) = \overline{tr(A)}$.
- $tr(A^*A) = 0$ iff $tr(AA^*) = 0$ iff $A = 0$.

  (Note: $tr(A^*A) = tr(AA^*) = \sum_{i=1}^{n}\sum_{j=1}^{n} |a_{ij}|^2$.)
- $tr(A + B) = tr(A) + tr(B)$ for all $B \in \mathbb{F}^{n\times n}$.
- $tr(AB) = tr(BA)$ for all $B \in \mathbb{F}^{n\times n}$.

Minor and cofactor

Let $A = [a_{ij}] \in \mathbb{F}^{n \times n}$ be a square matrix.

- The sub-matrix of A obtained by deleting the ith row and the jth column is called the $(i,j)^{th}$-minor of A, and is denoted by A_{ij}. Note that $A_{ij} \in \mathbb{F}^{(n-1)\times(n-1)}$.
- $C_{ij}(A) := (-1)^{i+j} det(A_{ij})$ is called the $(i,j)^{th}$-cofactor. (Sometime, we simply write $C_{ij} = C_{ij}(A)$.)
- The matrix $C = [C_{ij}] \in \mathbb{F}^{n \times n}$ is called the cofactor matrix of A.
- The matrix $\text{adj}(A) := C^t$ is called adjugate of A.
- We write $A_i(x)$ for the matrix obtained from A by replacing its ith row by a row vector $x \in \mathbb{F}^{1 \times n}$.

Determinant of a matrix

Definition

The determinant of $A \in \mathbb{F}^{n\times n}$, denoted by $\det(A)$ or $|A|$, is defined inductively as follows:

- If $n = 1$, then $\det(A) := a_{11}$.
- If $n > 1$, then

$$\det(A) := \sum_{j=1}^{n} a_{1j} C_{1j} = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j}).$$

- **Example:** Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Then

$$\begin{aligned} det(A) &= (-1)^{1+1} a_{11} \det[a_{22}] + (-1)^{1+2} a_{12} \det[a_{21}] \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned}$$

Determinant- Example

- $\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$

$$= (-1)^{1+1} \times 1 \times \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} + (-1)^{1+2} \times 2 \times \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (-1)^{1+3} \times 3 \times \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}$$

$$= 1 \times \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}$$

$$= (3 \times 2 - 1 \times 1) - 2 \times (2 \times 2 - 1 \times 3) + 3 \times (2 \times 1 - 3 \times 3)$$

$$= 5 - 2 \times 1 + 3 \times (-7) = -18.$$

Determinant- Example

- The determinant of a lower-triangular matrix is the product of its diagonal entries.

  Proof:

$$\det\begin{bmatrix} a_{11} & & & & \\ a_{21} & a_{22} & & & \\ a_{31} & a_{32} & a_{33} & & \\ \vdots & & & \ddots & \\ a_{n1} & & \cdots & & a_{nn} \end{bmatrix} = a_{11} \det\begin{bmatrix} a_{22} & & & \\ a_{32} & a_{33} & & \\ \vdots & & \ddots & \\ a_{n2} & & \cdots & a_{nn} \end{bmatrix}$$

$$= a_{11}a_{22} \det\begin{bmatrix} a_{33} & & & \\ a_{43} & a_{44} & & \\ \vdots & & \ddots & \\ a_{n3} & & \cdots & a_{nn} \end{bmatrix}$$

$$= \cdots$$

$$= a_{11}a_{22}\cdots a_{nn}.$$

Properties of determinant

Let $A, B \in \mathbb{F}^{n\times n}, 0 \neq \alpha \in \mathbb{F}$ and $1 \leq i \neq j \leq n$.

(1) $det(E[i,j]A) = -det(A)$.

(2) $det(E_\alpha[i]A) = \alpha\ det(A)$.

(3) $det(E_\alpha[i,j]A) = det(A)$.

(4) $det(A^t) = det(A)$ and $det(A^*) = \overline{det(A)}$.

(5) $det(AB) = det(A)\ det(B)$.

(6) If some row (or column) of A is the zero vector, then $det(A) = 0$.

(7) Determinant of a triangular matrix is the product of its diagonal entries.

(8) A Adj(A) = Adj$(A)A = det(A)I_n$.

(9) A is invertible iff $det(A) \neq 0$.

(10) $det(A_i(x+y)) = det(A_i(x)) + det(A_i(y))$ for all $x, y \in \mathbb{F}^{1\times n}$.

Determinant using RREF

Consider the row operations:

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix} \xrightarrow{E_1[2,1], E_1[3,1], E_1[4,1]} B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & -1 & 1 & 2 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

$$\xrightarrow{E_1[3,2], E_1[4,2]} C = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & 4 \end{bmatrix} \xrightarrow{E_1[4,3]} D = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

Using property (3), we get $det(A) = det(B) = det(C) = det(D)$.
Further, using property (7) we get $det(D) = 8$.
Hence $det(A) = 8$.

Problems

1. For each $n > 2$, construct an $n \times n$ nonzero matrix where no row is a scalar multiple of another row but its determinant is 0.
2. Let $A_{i\star}$ with rows $A_{1\star}, A_{2\star}, \cdots, A_{n\star}$. Is it true that if $det(A) = 0$, then some row, say $A_{i\star}$, can be written as

$$A_{i\star} = \alpha_1 A_{1\star} + \cdots + \alpha_{i-1} A_{(i-1)\star} + \alpha_{i+1} A_{(i+1)\star} + \cdots + \alpha_n A_{n\star}$$

   for some scalars α_j? What about columns instead of rows?
3. Give examples of $A, B \in \mathbb{R}^{2\times 2}$ such that $det(A + B) \neq det(A) + det(B)$.

Invertible matrices and RREF

Recall: A square matrix is said to be invertible (or nonsingular) if it has an inverse, otherwise said to be non-invertible (or singular).

Theorem

The following conditions are equivalent for a **square** matrix $A \in \mathbb{F}^{n \times n}$:

1. A is invertible.
2. $\det(A) \neq 0$.
3. The RREF of A is I_n.
4. $\text{rank}(A) = n$.
5. A is a product of elementary row operation matrices.

Inverse of a matrix using RREF

- Suppose A is an invertible square matrix.
- If E is the product of elementary row operation matrices, which used to get the RREF of A, then $EA = I$. Since A is invertible we have $E = A^{-1}$.
- Note that

$$E[A \,|\, I] = [EA_{\star 1}\ EA_{\star 2}\ \cdots\ EA_{\star n} \mid Ee_1\ Ee_2\ \cdots\ Ee_n] = [EA \,|\, EI] = [I \,|\, A^{-1}]$$

Procedure to compute the inverse of a matrix

- Start with the augmented matrix $[A \,|\, I]$. Apply elementary row operations to reduce A to its row reduced echelon form I, while simultaneously applying the operations on the entries of I.
- If A is invertible, then at the end these row-operations we get a matrix of the form $[I \,|\, B]$. From the above discussions, we conclude that $A^{-1} = B$.

Problems I

1. Find inverse of the following matrices.

   1.1 $\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ Ans: $\begin{bmatrix} 0.4 & -0.1 \\ -0.2 & 0.3 \end{bmatrix}$

   1.2 $\begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ Ans: $\begin{bmatrix} -0.7 & 0.2 & 0.3 \\ -1.3 & -0.2 & 0.7 \\ 0.8 & 0.2 & -0.2 \end{bmatrix}$

2. Let $A, B \in \mathbb{F}^{n \times n}$ satisfying $AB = I$. Show that both A, B are invertible and $BA = I$.

   (Not true for non-square matrices. For eg: $A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}$.)

Problems II

3. Let $A = \begin{bmatrix} 2 & 1 \\ 6 & 4 \end{bmatrix}$. Express A and A^{-1} as products of elementary matrices.
4. Let $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$. Find matrices X, Y such that $AX = B$ and $YA = B$. (Hint: Both A, B are invertible.)
5. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -b & -c \end{bmatrix}$ where $b, c \in \mathbb{C}$. Show that $A^{-1} = bI + cA$.
6. Show that if A is an upper-triangular invertible matrix, then so is A^{-1}.